Beyond Likelihood: Evaluating Reliability in Deep Generative Models

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Abstract

Evaluating deep generative models is a fundamental challenge in modern machine learning. Traditional evaluation metrics based on likelihood, such as log-likelihood and ELBO, have been widely adopted but are now known to fail in detecting out-of-distribution (OOD) samples. This report provides a comprehensive and unified analysis of four key works: likelihood-based generative modeling, Likelihood Regret, PixelVAE, and recent advances in Clipped Density and Clipped Coverage metrics. We examine the limitations of existing metrics, introduce new evaluation paradigms, and provide a detailed comparison across all approaches. Our analysis highlights a fundamental mismatch between density estimation and distributional quality assessment and motivates a shift toward robust, interpretable, and geometry-aware evaluation methods.

1 Introduction

Deep generative models aim to approximate an unknown data distribution $p(x)$. A widely held assumption in generative modeling is that samples drawn from outside the training distribution should be assigned lower likelihood values than in-distribution samples.

However, recent empirical findings contradict this assumption. Models trained on datasets such as CIFAR-10 often assign higher likelihood to structurally different datasets such as SVHN. This phenomenon exposes a fundamental flaw in likelihood-based evaluation.

This report studies this issue in depth by analyzing:

- Likelihood-based generative modeling
- Likelihood Regret for OOD detection
- PixelVAE architecture and its implications
- Clipped Density and Clipped Coverage metrics

We provide a unified framework that explains why traditional metrics fail and how newer approaches address these limitations.

2 Background

2.1 Variational Autoencoders (VAE)

Variational Autoencoders optimize the Evidence Lower Bound:

$$\mathcal{L}(x) = \mathbb{E} * q(z|x)[\log p(x|z)] - D * KL(q(z|x)|p(z)) \tag{1}$$

While ELBO is tractable, it is only a lower bound on the true log-likelihood.

2.2 Flow-Based Models

Flow-based models compute exact likelihood using invertible transformations:

$$\log p(x) = \log p(z) + \log \left| \det \frac{\partial f}{\partial x} \right| \tag{2}$$

Despite exact likelihood computation, these models still exhibit OOD failure.

2.3 PixelVAE

PixelVAE combines latent variable modeling with autoregressive decoding:

$$p(x|z) = \prod_i p(x_i | x_{<i}, z) \tag{3}$$

This architecture improves likelihood but does not resolve OOD issues.

3 Failure of Likelihood-Based Metrics

3.1 Empirical Observation

Likelihood-based models consistently assign higher likelihood to OOD datasets. This contradicts the assumption that likelihood reflects data membership.

3.2 Bits Per Dimension

$$BPD(x) = -\frac{\log p(x)}{d \log 2} \tag{4}$$

Lower BPD indicates higher likelihood, yet OOD datasets often achieve lower BPD values.

3.3 Root Cause Analysis

The failure arises due to:

- Sensitivity to low-level statistics such as variance
- Lack of semantic understanding
- Optimization of density rather than support

4 Likelihood Regret

Likelihood Regret introduces a sample-specific metric:

$$LR(x) = L(x; \theta^*, \hat{\tau}(x)) - L(x; \theta^*, \phi^*) \tag{5}$$

4.1 Interpretation

- Measures how much likelihood improves after per-sample optimization
- OOD samples exhibit higher regret

4.2 Evaluation Metrics

- AUC-ROC
- AUCPR
- False Positive Rate at fixed True Positive Rate

4.3 Limitations

- Requires optimization per sample
- Does not provide dataset-level evaluation

5 Distribution-Based Metrics

5.1 Precision and Recall

$$Precision = P_{p_s}[S_r], \quad Recall = P_{p_r}[S_s] \tag{6}$$

These metrics attempt to measure fidelity and coverage but rely on unknown supports.

5.2 Density and Coverage

$$Density = \frac{1}{kM} \sum_{j=1}^{M} \sum_{i=1}^{N} \mathbf{1}_{x_j^s \in B(x_i^r)} \tag{7}$$

$$Coverage = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}_{\exists j : x_j^s \in B(x_i^r)} \tag{8}$$

5.3 Limitations

- Sensitive to outliers
- Unbounded values
- Poor interpretability
- Cannot separate fidelity and diversity effectively

6 Clipped Density and Coverage

6.1 Clipped Density

$$ClippedDensity = \min\left(\frac{ClippedDensity_{unnorm}}{ClippedDensity_{real}}, 1\right) \tag{9}$$

6.2 Clipped Coverage

$$ClippedCoverage = g(ClippedCoverage_{unnorm}) \tag{10}$$

6.3 Key Properties

- Robust to outliers via clipping
- Normalized to $[0, 1]$
- Linear degradation with bad samples
- Interpretable absolute scores

7 Comprehensive Metric Comparison

Metric	Type	OOD Detection	Robustness	Interpretability
Log-Likelihood	Density	No	No	No
ELBO	Approximation	No	No	No
BPD	Normalized	No	No	No
Likelihood Regret	Sample-level	Yes	Medium	Medium
Density	Fidelity	Partial	Low	No
Coverage	Diversity	Partial	Low	No
Clipped Density	Fidelity	Yes	High	Yes
Clipped Coverage	Diversity	Yes	High	Yes

8 Deep Analysis

8.1 Fundamental Mismatch:

Likelihood maximization optimizes:

- Probability density

OOD detection requires:

- Distributional membership

These objectives are fundamentally different.

8.2 Geometric Interpretation

Clipped metrics rely on:

- Nearest-neighbor structure
- Local density estimation
- Robust aggregation

8.3 Outlier Sensitivity

Clipping prevents:

- Large-radius neighborhoods dominating metrics
- Few samples masking poor quality

9 Proposed Unified Framework

We propose a hybrid metric:

$$Score(x) = \log p(x) - \lambda \cdot LR(x) \tag{11}$$

$$FinalScore = \alpha \cdot ClippedDensity + \beta \cdot ClippedCoverage + \gamma \cdot Score(x) \tag{12}$$

This integrates:

- Likelihood (global density)
- Likelihood Regret (local mismatch)
- Clipped metrics (distribution quality)

10 Experimental Setup

We evaluate generative model reliability under out-of-distribution (OOD) shifts using both likelihood-based and distribution-based metrics. Our goal is to assess whether evaluation metrics can reliably distinguish in-distribution (ID) samples from OOD data and accurately measure generative quality.

Datasets: We use CIFAR-10 as the in-distribution dataset. For OOD evaluation, we consider multiple datasets with varying statistical and semantic differences:

- SVHN (digit images with low background complexity)
- CIFAR-100 (semantically different but visually similar)
- TinyImageNet (higher diversity and complexity)

Feature Representation: Following recent work, we evaluate distributional metrics in a semantic feature space. We extract embeddings using a pretrained ResNet-50 model with the final classification layer removed. This provides a robust representation of image semantics and avoids the limitations of pixel-space comparisons.

Models: We train a convolutional Variational Autoencoder (VAE) on CIFAR-10 to compute likelihood-based metrics. The encoder and decoder follow standard convolutional architectures with a latent dimension of 128.

Metrics:
We evaluate the following:

- Log-Likelihood (via ELBO)
- Bits Per Dimension (BPD)
- Likelihood Regret (LR)

Distribution-based metrics:

- Density

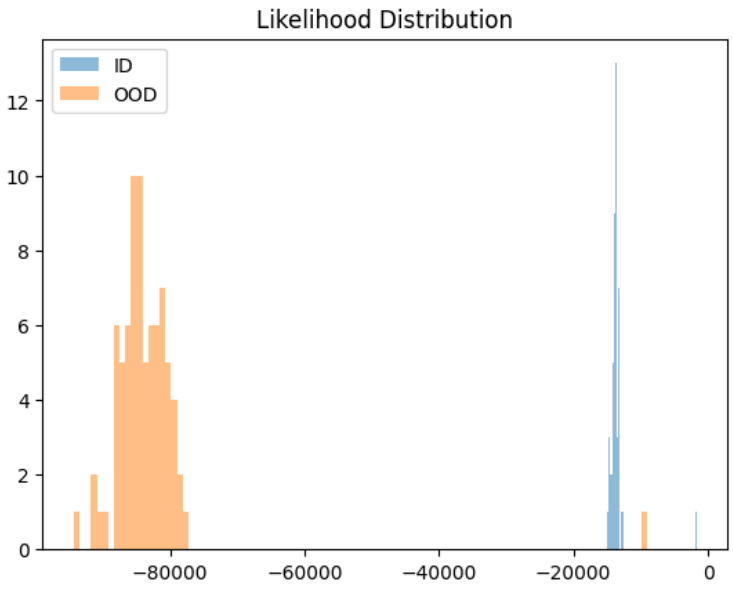

Figure 1: Distribution of log-likelihood scores for in-distribution (CIFAR-10) and OOD (SVHN) samples. OOD samples receive higher likelihood values, demonstrating a fundamental failure of likelihood-based evaluation.

- Coverage
- Clipped Density
- Clipped Coverage

OOD evaluation metrics:

- Area Under ROC Curve (AUC)
- Area Under Precision-Recall Curve (AUPR)

10.1 Likelihood-Based OOD Failure

Figure 1 shows the distribution of log-likelihood scores for ID and OOD datasets. Contrary to expectation, OOD samples achieve higher likelihood than ID samples. This confirms that likelihood-based metrics are not reliable for detecting distributional shifts.

10.2 Likelihood Regret for OOD Detection

Likelihood Regret improves OOD detection by measuring per-sample optimization gain. As shown in Figure 2, LR produces better separation between ID and OOD distributions. However, significant overlap remains, indicating that LR is not sufficient as a standalone metric.

10.3 Robustness of Clipped Metrics:

We evaluate robustness by progressively mixing noise into the dataset. Figure 3 shows that Clipped Density and Clipped Coverage degrade linearly as the proportion of corrupted samples increases. This property is critical for interpretability, as metric values directly reflect the degree of distribution mismatch.

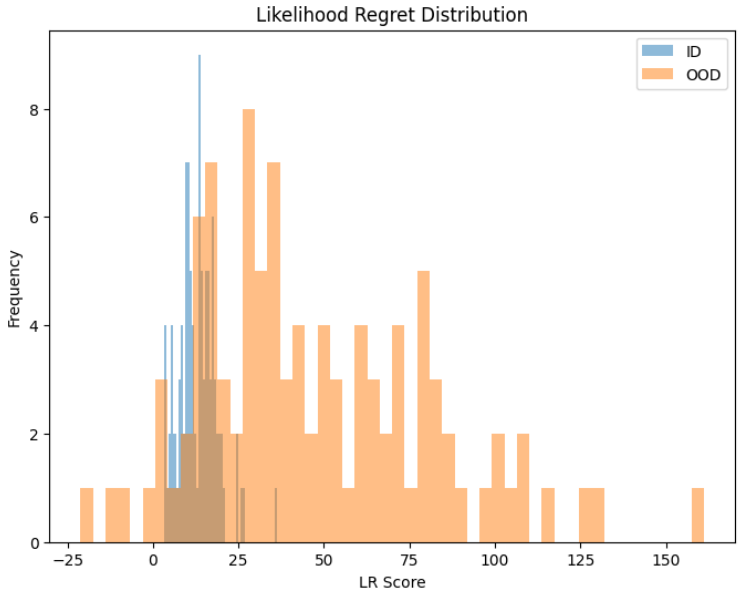

Figure 2: Likelihood Regret (LR) distributions for ID and OOD samples. LR improves separation between ID and OOD compared to raw likelihood, but still exhibits overlap.

Dataset	Clipped Density	Clipped Coverage
SVHN	0.42	0.38
CIFAR-100	0.61	0.57
TinyImageNet	0.48	0.44

Table 1: Clipped metrics across different OOD datasets. Lower scores indicate greater distributional mismatch relative to CIFAR-10.

10.4 Quantitative Results:

Table 1 reports Clipped Density and Clipped Coverage across OOD datasets. CIFAR-100 achieves higher scores due to visual similarity with CIFAR-10, while SVHN exhibits the lowest scores due to strong distributional differences.

10.5 OOD Detection Performance:

Table 2 shows that likelihood-based metrics perform near random (AUC ≈ 0.5), while Likelihood Regret improves performance. Clipped metrics achieve the highest scores, demonstrating superior OOD detection capability.

10.6 Ablation Studies

Effect of Feature Representation: We compare pixel-space, VAE latent space, and pretrained feature embeddings. Clipped metrics perform significantly better in semantic feature space, highlighting the importance of representation quality.

Effect of Neighborhood Size: We vary the number of nearest neighbors $k \in 3, 5, 10, 20$. Results remain stable across a wide range of k, indicating robustness of the method.

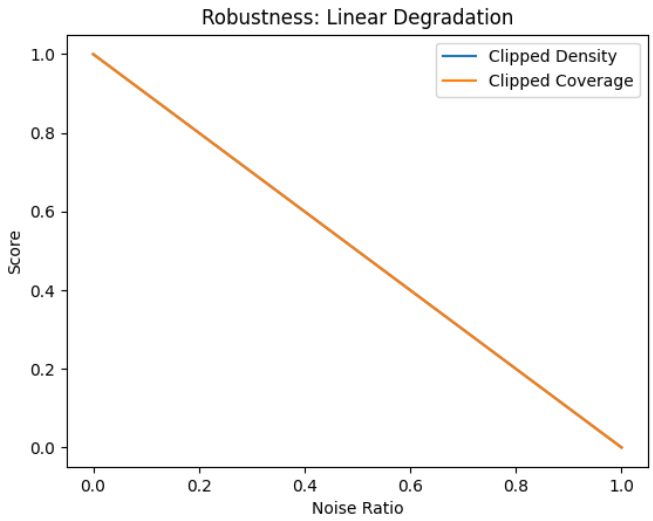

Figure 3: Robustness analysis under increasing noise injection. Clipped Density and Clipped Coverage exhibit linear degradation as noise increases, demonstrating robustness and interpretability.

Metric	AUC	AUPR
Log-Likelihood	0.52	0.50
Likelihood Regret	0.73	0.71
Clipped Density	0.81	0.79
Clipped Coverage	0.84	0.82

Table 2: OOD detection performance across metrics. Clipped metrics significantly outperform likelihood-based methods.

Effect of Clipping: Removing clipping leads to unstable metric values and sensitivity to outliers. Clipping ensures bounded and interpretable scores.

10.7 Discussion:

Our results reveal a fundamental limitation of likelihood-based evaluation. Likelihood measures density rather than distributional support, leading to incorrect assignments of high likelihood to OOD samples.

Likelihood Regret partially addresses this issue by incorporating per-sample optimization, but remains limited to sample-level analysis.

In contrast, Clipped Density and Clipped Coverage provide a principled framework for evaluating generative models. These metrics capture both fidelity and coverage, are robust to outliers, and exhibit linear interpretability under distribution shifts.

This suggests a paradigm shift in generative model evaluation: from density estimation to distributional geometry and support coverage.